MTH 201 Chapter 5 Infinite Series

Series Review 2 (covers 5.2 – 5.6)

Directions: Determine if each of the following series converges or diverges. Use any method.

1. $\sum_{k=1}^{\infty}\left(\frac{k+2}{3k-1}\right)^k$

2. $\sum_{k=1}^{\infty}\frac{3^k k^2}{k!}$

3. $\sum_{k=1}^{\infty}\left[\tan^{-1}(k+1)-\tan^{-1}(k)\right]$

4. $\displaystyle\sum_{k=1}^{\infty}\frac{k^{4/3}}{8k^2+5k+1}$

5. $\displaystyle\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k^2+1}$

6. $\displaystyle\sum_{k=1}^{\infty}\frac{(2)^{k+1}}{2k}$

7. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k(k+2)}$

8. Classify each of the following series as absolutely convergent, conditionally convergent, or divergent.

a) $\displaystyle\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{10k}$

b) $\displaystyle\sum_{k=1}^{\infty} \frac{\sin(k)}{k^3}$

c) $\displaystyle\sum_{k=1}^{\infty} \frac{(-1)^{k+1} 3^{2k-1}}{k^2+1}$